VentureMiner AI

Research Pipeline

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The research pipeline is the per-dimension work that turns a candidate into a validated opportunity. This document specifies the depth levels, the per-dimension work, and the verification step.

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1. Purpose & Scope

This document is the contract for the validation pipeline — the system that takes a candidate opportunity and produces a structured, cited set of evidence across the eight standard dimensions.

2. Depth levels

Quick	1	< 60s	50k	Triage, scanning many
Standard	3	< 8 min	400k	Default
Deep	5+	< 30 min	1.2M	High-stakes (board, investor)

3. Dimensions

The eight standard dimensions:

1. Market — size, growth, segmentation (AGT-RSRCH-MARKET).
2. Demand — search/social/intent signals (AGT-RSRCH-DEMAND).
3. Competitive — map, gaps (AGT-RSRCH-COMP).
4. Pricing — competitor pricing, WTP band (AGT-RSRCH-PRICING, AGT-RSRCH-WTP).
5. Persona — buyer personas (AGT-RSRCH-PERSONA).

8. Adjacency — relation to user's existing business (custom rubric, AGT-RSRCH-COMP).

4. Per-dimension flow

```
plan → retrieve (RAG) → fetch (plugins) → synthesize → self-check
```

Detailed per agent in Document 09.

5. Cross-dimension coherence

Some claims span dimensions (e.g. WTP depends on persona and pricing). The pipeline:

- Identifies cross-dimension dependencies at plan time.
- Sequences the dependent dimensions so the upstream is finished first.
- Re-checks downstream claims when upstream changes (incremental re-validation).

6. Evidence model

```
class Evidence(BaseModel):
claim: str
citations: list[Citation]
freshness: Freshness
confidence: Confidence
snippet: str
source_url: str
captured_at: datetime
agent_id: str
step_id: UUID
```

Every claim is bound to ≥ 1 citation. The verifier audits this discipline.

7. Verification

Every dimension's output is verified (AGT-VERIFY):

- Each claim has a citation.
- Each citation is real (the chunk exists and the snippet matches).
- The claim is consistent with the source.
- No policy violations.

If verification fails, the orchestrator retries with corrective feedback. Two consecutive failures mark the dimension unverified and surface to the user.

8. Human-in-the-loop

The user can:

- Attach manual evidence to a dimension (uploaded doc, link, or note).

9. Cost & latency

See Document 08 §9.

10. Failure modes

Source unavailable	Degrade; mark partial; surface in report.
RAG empty for a claim	Mark unverified; suggest user-supplied evidence.
LLM provider error	Retry; fallback; surface.
Verifier rejects twice	Mark unverified; surface.
Token budget exceeded	Skip non-critical dimensions; surface.
Wall-clock exceeded	Cancel; persist partial.

11. Appendix

11.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

11.2 Cross-references

- Application Flow: Document 03 §5.
- Multi-Agent: Document 08.
- Agent Specs: Document 09.
- RAG: Document 10.
- Scoring Engine: Document 16.

End of Document 15 — Research Pipeline. The verification step is the trust backbone of the entire system.